

18 — Mentor Q&A Bank (30+ Predicted Questions & Technical Answers)

This document provides answers to 30+ technical and operational questions that a project mentor or evaluation panel may ask during an oral examination.

■ Section 1: Basic & Overview Questions

Q1: What is the main objective of your project?

- **Simple Answer:** To help maritime ports predict upcoming cargo volume and manage port logistics using machine learning.
 - **Technical Answer:** The primary objective is to build a data-grounded decision support platform that ingests multi-year historical port traffic datasets, fits Holt-Winters + Ridge time-series ML models, and outputs commodity-wise volume forecasts with 95% dynamic prediction bounds.
 - **If mentor asks deeper** →: Point them to **CODE_PH_0** and explain how **CODE_PH_1** is aggregated across 16+ commodities.
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Q2: Who are the target end-users of this platform?

- **Simple Answer:** Port managers, harbor masters, customs officers, and terminal logistics operators.
 - **Technical Answer:** The system serves operations executives requiring berth utilization metrics (**CODE_PH_0**), harbor masters tracking AIS vessel delay probabilities (**CODE_PH_1**), and logistics managers running What-If scenario simulations (**CODE_PH_2**).
 - **If mentor asks deeper** →: Explain how different tabs tailor data for different roles (Tab 1 for C-suite executives, Tab 3 for logistics planners, Tab 5 for risk managers).
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Q3: Why did you focus on Cargo rather than just Vessel tracking?

- **Simple Answer:** Because ports generate revenue and manage logistics based on cargo weight, not just ship counts.
 - **Technical Answer:** Vessel counts alone do not indicate resource requirements. A 300-meter crude tanker requires oil jetties and pipelines, while a container vessel requires gantry cranes and yard stacking. Focusing on cargo tonnage enables precise terminal infrastructure and draft depth mapping.
 - **If mentor asks deeper** →: Show how **CODE_PH_0** in **CODE_PH_1** maps commodities to specific draft depths and hinterland consumers.
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■■ Section 2: Architecture & Technical Stack Questions

Q4: Why did you choose FastAPI over Flask or Django for the backend?

- **Simple Answer:** FastAPI is faster, handles modern asynchronous requests, and generates interactive API documentation automatically.
 - **Technical Answer:** FastAPI is built on Starlette and Pydantic, executing asynchronous ASGI handlers with low overhead. It performs automatic request validation and generates interactive OpenAPI Swagger UI at **CODE_PH_0**.
 - **If mentor asks deeper** →: Show **CODE_PH_0** where routers are mounted with clean dependency injection.
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Q5: Why did you select Streamlit for the frontend?

- **Simple Answer:** Streamlit allows us to build an interactive data science web UI directly in Python without complex JavaScript frameworks.
 - **Technical Answer:** Streamlit provides rapid UI state binding, native dataframe widgets, and seamless integration with Plotly charts, letting us focus on analytical pipeline logic.
 - **If mentor asks deeper** →: Explain how custom CSS injections in **CODE_PH_0** transform default Streamlit styles into a custom design system.
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Q6: How do the frontend and backend communicate?

- **Simple Answer:** The frontend sends HTTP requests to backend REST API endpoints on port 8000 and receives JSON data.
 - **Technical Answer:** **CODE_PH_0** uses helper functions **CODE_PH_1** and **CODE_PH_2** powered by Python's **CODE_PH_3** library to issue REST queries to **CODE_PH_4**. Responses are parsed from JSON into Streamlit session state and Plotly render objects.
 - **If mentor asks deeper** →: Trace **CODE_PH_0** from **CODE_PH_1** to **CODE_PH_2**.
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■ ■ Section 3: Backend & Data Questions

Q7: Where is the dataset stored, and how is it loaded?

- **Simple Answer:** The data is stored in **CODE_PH_0** and loaded into memory using Pandas.
 - **Technical Answer:** The system reads **CODE_PH_0** (1,354 records) on demand using **CODE_PH_1**. Function **CODE_PH_2** processes the dataset into an in-memory time-series DataFrame.
 - **If mentor asks deeper** →: Point to **CODE_PH_0** to show how the backend also supports dynamic multipart CSV file uploads (**CODE_PH_1**).
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Q8: How does the backend handle missing data or missing months in the dataset?

- **Simple Answer:** It fills missing commodity-month pairs with zero volume so time series remain continuous.

- **Technical Answer:** `CODE_PH_0` constructs a complete month-by-month grid from `CODE_PH_1` to `CODE_PH_2` using Pandas `CODE_PH_3`. Missing commodity observations are zero-filled (`CODE_PH_4`) to maintain equal step intervals required for lag calculations.
 - **If mentor asks deeper** →: Explain why zero-filling is mathematically safer than dropping rows when calculating lag 12 (y_{t-12}) seasonal features.
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Q9: Does the system use a SQL database?

- **Simple Answer:** Not in the PoC build; it uses an in-memory Pandas data pool backed by CSV persistence.
 - **Technical Answer:** To ensure high-speed analytical calculations without external database overhead, data is maintained in Pandas DataFrames. However, the system exposes dynamic CSV upload endpoints (`CODE_PH_0`) and can easily swap the Pandas reader for SQLAlchemy/PostgreSQL.
 - **If mentor asks deeper** →: Reference Tab 9 (`CODE_PH_0`), which shows simulated architecture statistics for PostgreSQL and TimescaleDB pools.
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■ Section 4: AI & ML Questions

Q10: What machine learning algorithms are used for cargo forecasting?

- **Simple Answer:** We use Holt-Winters Exponential Smoothing combined with Ridge Regression.
 - **Technical Answer:** The primary model applies Holt-Winters Exponential Smoothing to decompose level and seasonal trends, combined with Multi-Feature Ridge Regression with L2 regularization to evaluate lagged features ($t-1$, $t-12$) and vessel arrival counts.
 - **If mentor asks deeper** →: Show `CODE_PH_0` where model predictions and residual error scaling generate the 95% dynamic prediction interval.
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Q11: Why didn't you use deep learning models like LSTM or Transformer for forecasting?

- **Simple Answer:** Deep learning requires millions of data points and tends to overfit small monthly tabular datasets.
 - **Technical Answer:** Monthly port datasets (1,354 records) are structured time-series data. LSTMs contain millions of parameters and overfit on small sample sizes. Ridge Regression with L2 regularization guarantees stable, explainable weights with zero overfitting.
 - **If mentor asks deeper** →: Explain that deep learning models (LSTM/Autoencoder) are used in `CODE_PH_0` for anomaly detection where sequence pattern matching is required.
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Q12: How do you evaluate the accuracy of your forecasting model?

- **Simple Answer:** We compare our model against a Seasonal Naive baseline using WAPE, MAPE, MAE, and RMSE metrics.

- **Technical Answer:** `CODE_PH_0` performs chronological out-of-sample backtesting against actual test data, calculating Weighted Absolute Percentage Error (WAPE), MAPE, MAE, and RMSE.

* If mentor asks deeper →: Write out the WAPE formula:
$$\text{WAPE} = \frac{\sum |y - \hat{y}|}{\sum y} \times 100$$

$y - \hat{y}$

$\sum y \times 100$ and explain why it avoids division-by-zero artifacts.

Q13: What are "Explainable Forecast Drivers"?

- **Simple Answer:** It shows the exact percentage contribution of real-world factors causing cargo volume to rise or fall.
- **Technical Answer:** It is a regression weight decomposition algorithm that calculates relative feature importance for factors like hinterland industrial demand (MRPL/UPCL), monsoon swells, and trade lane velocity.
- **If mentor asks deeper** →: Point to `CODE_PH_0` and show how driver weights sum to 100%.

Q14: How does the What-If Cargo Scenario Simulator work?

- **Simple Answer:** It allows users to move sliders for vessel arrivals, demand, and weather to see the net impact on future cargo volume.
- **Technical Answer:** `CODE_PH_0` accepts parameter offsets (`CODE_PH_1`, `CODE_PH_2`, `CODE_PH_3`), recalculates the time-series vector, computes simulated throughput, and categorizes capacity risk levels.
- **If mentor asks deeper** →: Show `CODE_PH_0` handling POST requests from Streamlit Tab 3.

■ Section 5: Cargo & Domain Specific Questions

Q15: Which port's data is used in this project?

- **Simple Answer:** New Mangalore Port Authority (NMPA) in Karnataka, India.
- **Technical Answer:** The primary dataset `CODE_PH_0` contains actual traffic records for NMPA, covering major terminals like Oil Jetties, JSW Container Terminal, UPCL Coal Berths, and KIOCL Iron Ore Jetty.
- **If mentor asks deeper** →: Point to `CODE_PH_0` in `CODE_PH_1` line 35.

Q16: How does the platform distinguish between Imports and Exports?

- **Simple Answer:** By filtering the `CODE_PH_0` column into `CODE_PH_1` (Exports) and `CODE_PH_2` (Imports).
- **Technical Answer:** The dataset contains explicit `CODE_PH_0` tags. `CODE_PH_1` represents outward export shipments (e.g., POL products, iron ore pellets), while `CODE_PH_2` represents inward imports (e.g., crude oil, thermal coal, fertilizers).
- **If mentor asks deeper** →: Show the `CODE_PH_0` dropdown in Streamlit Tab 3.

Q17: What commodities are analyzed in the system?

- **Simple Answer:** 16 major commodities including Crude Oil, Thermal Coal, Containers, Iron Ore, Fertilizers, LPG, POL Products, Edible Oil, and Cement.
 - **Technical Answer:** The system categorizes cargo into 16 NMPA commodity strings extracted dynamically via `CODE_PH_0`.
 - **If mentor asks deeper** →: Explain how `CODE_PH_0` in `CODE_PH_1` assigns distinct color tokens to each commodity for consistent UI visualization.
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■ Section 6: Old vs. New & Scalability Questions

Q18: What is the single biggest improvement in the current version over the previous version?

- **Simple Answer:** Shifting the core focus from basic ship counting to **Predictive Cargo Analytics** backed by backtested ML models.
 - **Technical Answer:** The previous version was a passive vessel tracker displaying static ship locations. The new version introduces a time-series ML forecasting engine, chronological WAPE backtesting, explainable drivers, and interactive scenario simulators.
 - **If mentor asks deeper** →: Point to `CODE_PH_0`.
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Q19: Is the system scalable to handle other major ports like JNPT or Mundra?

- **Simple Answer:** Yes, because the data pipeline and ML models are commodity-agnostic.
 - **Technical Answer:** The architecture separates data ingestion from model execution. Adding JNPT or Mundra simply requires passing a new CSV file to `CODE_PH_0` or uploading it via `CODE_PH_1`.
 - **If mentor asks deeper** →: Reference `CODE_PH_0`, which already contains benchmark data for JNPT, Mumbai, Chennai, Kandla, Vizag, and Kochi.
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Q20: How does the AI Maritime Copilot work?

- **Simple Answer:** It parses natural language questions using a 3-tier fallback execution model.
- **Technical Answer:** `CODE_PH_0` implements a LangGraph cognitive dispatcher. Tier 1 handles complex multi-variable queries via LLM synthesis, Tier 2 uses SLM fast reasoning, and Tier 3 executes deterministic zero-LLM rule matching.
- **If mentor asks deeper** →: Show the step-by-step dispatch trace rendered on Streamlit Tab 8.